# Matthew Ngoy

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## Summary

Computer science senior applying data engineering, security testing and systems analysis.

## Education

**Kean University** Union, NJ

*B.S. Computer Science (Cybersecurity Option); GPA: 3.52* *Expected May 2027*

- **Completed Coursework:** Data Structures; Algorithms; Operating Systems; Databases; Computer Security; Cryptography; Ethical Hacking; Digital Forensics.
- **Honors & Programs:** NSF IUSE Scholar (2025, 2026); CRA UR2PhD; CAHSI Travel Scholar; Dean's List; Upsilon Pi Epsilon; WELL Program; Young Envoys Scholarship.

## Technical Skills

**Languages:** Python, Java, JavaScript, SQL, PHP, PowerShell
**Cybersecurity:** OSINT, Nmap, adversarial testing, OWASP LLM Top 10, MITRE ATLAS
**Artificial Intelligence & LLM Evaluation:** LLM-as-judge, OpenAI API, Ollama
**Systems & Development:** Linux (Ubuntu, Kali), Git, Docker, MySQL, SQLite, Node.js, Arduino/ESP8266

## Experience

**Cybersecurity Research Intern** Sep. 2026 – Present

*Kean University Center for Cybersecurity* *Union, NJ*

- Support solar-power integration for Mobile ACDC, a portable teaching data center, by defining rack-load, battery-capacity and power-monitoring requirements.
- Explained phishing, public Wi-Fi risks, password security and digital privacy as a student panelist on WKNJ 90.3 FM's C.Y.B.E.R. Talk.

**Cybersecurity Intern (WELL Program)** May – Aug. 2026

*Kean University Center for Cybersecurity* *Union, NJ*

- Classified nonprofit cyber incidents using OSINT, confidentiality, integrity and availability (CIA) impact, CISA sectors and TAG Cyber categories.
- Integrated browser requests, ESP8266 Wi-Fi and Arduino MEGA I/O for smart-home control.
- Prepared Port Authority of NY/NJ STEaM Camp teaching environments on 30 laptops.
- Taught cryptography, programming and IP addressing in 12 lessons for 60 high school students.

## Projects

**CougarCTF 2027** | *Python, Docker, CTFd, GitHub Actions* Sep. 2026 – Present

- Co-develop a 200-challenge board with two teammates, requiring cryptanalytic, forensic and execution-state reasoning.
- Enforced complete solver execution in continuous integration (CI); corrected a flag leak and 10 hint/text issues from player-view review.

**LLM Red-Team Harness** | *Python, SQLite, Ollama* June 2026

- Compared canary checks and rubric judges for instruction adherence across 28 probes and 2 local models.
- Questioned defense comparisons; subsequent re-analysis distinguished filtered-output safety from raw model adherence.

## Research & Presentations

**HadAgent Verification Study** | *Advisor: Dr. Boyang Li* Sep. 2026 – Present

- Identified how untrusted score-scale metadata altered an inference verifier's threshold decisions; investigated execution-evidence limits and disclosed validation gaps to the authors.

**Gen Z Housing Demographics** | *NSF IUSE Scholars Academy* Apr. – Sep. 2026

- Analyzed 3,114 counties using compound geographic keys, population weights and rent-specific missing-data rules.
- Connected SQL/PHP/JavaScript for county drill-down; presented the weak county-level association between young-adult concentration and rent as sole student author at Great Minds in STEM.

**Debate Coach LLM Evaluation** | *CRA UR2PhD* Feb. – Apr. 2026

- Contributed to fine-tuning debate coaches with two teammates; refined tone and complexity to support constructive dissent through argument rubrics, citation checks and user interviews.

## Leadership

**Secretary** Sep. 2026 – Present

*Women in CyberSecurity (WiCyS), Kean University Chapter* *Union, NJ*

- Coordinated PHD4FREE!; co-presented research and funding pathways with PhD student Demetrius Hernandez to 15 attendees.
- Plan WiCyS CTF outreach to universities, companies and prospective speakers.

**Peer Mentor / Code Samurai** Sep. 2026 – Present; Jan. – May 2025

*Kean University Department of Computer Science and Technology* *Union, NJ*

- Mentor students in cybersecurity, Python, SQL and data structures through debugging and stepwise explanation.

**International Youth Delegate / IFWY Fellow** Sep. – Oct. 2025

*International Forum We, the Youth (IFWY)* *New York, NY / Seoul, South Korea*

- Proposed SelectSolar to track solar-panel condition and inform reuse or recycling decisions; received 97 public upvotes.
- Recommended extended producer responsibility at the fully funded Cornell Tech dialogue; peer-elected to continue policy work in Seoul as 1 of 25 North American delegates.